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Herbal medicine Gamgungtang down-regulates autoimmunity through induction of TH2 cytokine production by lymphocytes in experimental thyroiditis model

Eun-Ho Sa[a,b], Un-Ho Jin[a], Dong-Soo Kim[c], Bong-Seok Kang[a,1], Ki-Tae Ha[b], June-Ki Kim[b], Won-Hwan Park[b], Cheorl-Ho Kim[a,*]

[a] *Department of Biological Science, Sungkyunkwan University, Chunchun-Dong 300, Jangan-Gu, Suwon City, Kyunggi-Do 440-746, Republic of Korea*
[b] *Department of Oriental Medicine, Dongguk University and Cardiovascular MRC, Kyungju 780-714, Republic of Korea*
[c] *Department of Food Science and Technology, Kyungsung University, Nam-Gu, Busan 608-736, Republic of Korea*

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Abstract

The crude herbal formulation, Gamgungtang (GGT), has been shown to protect animals against a wide range of spontaneously developing or induced autoimmune diseases. We have previously reported that GGT shows marked down-regulation of several experimental autoimmune diseases. Although very effective at preventing thyroid infiltrates in mice immunized with mouse deglycosylated thyroglobulin and complete Freund's adjuvant and in spontaneous models of thyroiditis, it completely failed to modify experimental autoimmune thyroiditis (EAT) induced in mice immunized with mouse thyroglobulin and lipopolysaccharide. In this study, in an effort to elucidate the mechanisms by which GGT suppresses EAT, and autoimmunity in general, we investigated the in vivo effects of this drug on the Th1/Th2 lymphocyte balance, which is important for the induction or inhibition of autoreactivity. Naive SJL/J mice were treated orally for 5 days with GGT (80 mg/(kg day)). Spleen cells were obtained at various time points during the treatment period and were stimulated in vitro with concanavalin A. Interleukins IL-4, IL-10 and IL-12, transforming growth factor-β (TGF-β) and interferon-γ (IFN-γ) cytokine production was evaluated at the protein levels of the cytokines in the medium and mRNA expressions. A significant upregulation of IL-4, IL-10 and TGF-β was observed following treatment with GGT, which peaked at day 5 (IL-10) or day 10 (IL-4). On the other hand, IL-12 and IFN-γ production were either unchanged or decreased. It seems therefore that GGT induces in vivo a shift towards Th2 lymphocytes which may be one of the mechanisms of down-regulation of the autoimmune reactivity in EAT. Our observations indicate that down-regulation of TH1 cytokines (especially IL-12) and enhancement of Th2 cytokine production may play an important role in the control of T-cell-mediated autoimmunity. These data may contribute to the design of new immunomodulating treatments for a group of autoimmune diseases.

Keywords: Experimental autoimmune thyroiditis; Immunomodulation; TH1; TH2; Cytotoxic T lymphocyte; In vivo animal model; Gamgungtang; Glycyrrhizae radix (*Glyrrhiza uralensis* Fischer; Glycyrrhiza Fissch); Black bean (*Glycine max* L. Merr); Angelicae radix (*Angelica gigas* Nakai); Cnidii rhizoma (*Cnidium officinale* Makino)

1. Introduction

Autoimmune thyroid disease affects up to 1% of the population and is characterized by the presence of infiltrating lymphocytes and circulating antibodies to thyroid antigens. Thyrocyte destruction and thyroid atrophy are obvious features of autoimmune thyroid disease, occurring in both human and experimental thyroiditis, but the mechanisms responsible are not fully understood. A number of observations suggest a crucial participation of lymphokines in pathology of autoimmune thyroiditis. It was shown that mitogen-stimulated T-cell clones

Abbreviations: GGT, Gamgungtang; HT, Hashimoto's thyroiditis; IFN, interferon; TNF, tumor necrosis factor; MHC, major histocompatibility complex; CTL, cytotoxic T lymphocyte; EAT, experimental autoimmune thyroiditis; Tg, thyroglobulin; Endo-F, endo-β-*N*-acetylglucosaminidase F; TMSF, trifluoromethanesulfonic acid; pTg, porcine thyroglobulin; dgpTg, deglycosylated porcine Tg; Tc1, type 1 cytotoxic T-cell; p.i., post-immunization

* Corresponding author. Tel.: +82 31 290 7002; fax: +82 31 290 7015.
E-mail address: chkimbio@skku.edu (C.-H. Kim).
[1] Present address: Hormel Institute, University of Minnesota, Austin, MN 55912, USA.

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derived from thyroid glands from Hashimoto's thyroiditis (HT) patients secrete predominantly interferon (IFN)-γ (Del Prete et al., 1987) and tumor necrosis factor (TNF)-α (Del Prete et al., 1989; Turner et al., 1987). Another recently described feature of autoimmune thyroid disease is an increased frequency of major histocompatibility complex (MHC) class II antigens on some of the thyroid cells in the affected gland (Hanafusa et al., 1983). This aberrant expression can render the thyroid follicular cell capable of presenting autoantigens to the immune system (Hanafusa et al., 1983; Bottazzo et al., 1983) and initiate or perpetuate the autoimmune response.

There is a spectrum of human autoimmune thyroiditis in which antibodies and T-cells contribute differentially to the pathology (Dawe et al., 1993). A variety of animal models for experimental autoimmune thyroiditis (EAT) exist which may arise as a result of different mechanisms. One model extensively employed involves the immunization of susceptible mice (H-2^k) with autologous thyroglobulin in complete Freund's adjuvant (CFA) or lipopolysaccharide (LPS) (Hutchings et al., 1992; Esquivel et al., 1977). In this model, mononuclear cells consisting of both $CD4^+$ and $CD8^+$ T-cells, together with macrophages and a few B cells (Flynn et al., 1989), infiltrate the gland. There has been much speculation on the relative contribution of T helper type 1 (Th1) and Th2 subsets in this model and there is some evidence that the EAT induced in normal CBA/J (I-A^k) is characteristic of a Th1 response (Damotte et al., 1997). However, It was also reported that the Th2 subset may be involved, as in their model of granulomatous EAT, inhibiting the Th1 subset with anti-interferon-γ (IFN-γ) antibody did not prevent thyroid infiltration (Stull et al., 1992). Our recent study have also evaluated the contribution of these $CD8^+$ CTL cells to the pathogenesis of EAT by immunization of naive CBA/J mice with *N*-glycanase F and trifluoromethanesulfonic acid (TMSF)-deglycosylated porcine thyroglobulin (dgpTg) without the aid of adjuvants. dgpTg-immunized mice developed a different EAT from conventional EAT induced by thyroglobulin (Tg) and adjuvants; i.e., cytotoxic responses to dgpTg were detected in T-cells from dgpTg-immunized mice (Kim et al., 2004a,b).

Another murine model of thyroiditis has recently been developed which may be more akin to the spontaneously arising human disease. Wicker and colleagues have derived a non-obese diabetic (NOD) congenic mouse (NOD H-2^{h4}) bearing I-A^k, which spontaneously develops EAT but no insulin-dependent diabetes mellitus (IDDM). This EAT can be accelerated and enhanced by the addition of iodine to the drinking water (Podolin et al., 1993; Rasooly et al., 1996). The relative importance of Th1 and Th2 subsets in this model has not been established. Rose and colleagues found the anti-mouse thyroglobulin antibodies in this mouse model were predominantly immunoglobulin G2a (IgG2a) and IgG2b with no IgG1 (Rasooly et al., 1996), whereas Damotte et al., using an independently derived cross of NOD and B10BR mice, found IgG2a poorly represented in these animals (Damotte et al., 1997). The NOD mouse can therefore be considered a model for experimental thyroiditis (Goillot et al., 1991).

Gamgungtang (GGT), which consisted of four medicinal herbs such as Glycyrrhizae Radix (*Glyrrhiza uralensis* Fischer; Glycyrrhiza Fissch), Black bean (*Glycine max* L. Merr), Angelicae radix (*Angelica gigas* Nakai) and Cnidii Rhizoma (*Cnidium officinale* Makino), has been shown to stimulate several immune functions and has been found to be effective in the prevention of cancer metastases (Cho et al., 1999). It also modifies a number of animal models of autoimmune diseases, including experimental autoimmune encephalomyelitis (EAE) and collagen-induced arthritis (CIA) (Han et al., 1999). They also showed a strong cytotoxicity to human A549 lung carcinoma cells, cervix epithelioid carcinoma Hela cells and Hepa1c1c7 hepatoma cells with approximately 80% growth inhibition (Cho et al., 2000a,b). GGT increased the proliferation of T and B cells. In addition, GGT remarkably increased cytolytic activity of purified T-cell against K562 cells, indicating its immunomodulating effect [19]. Recently, it was shown that GGT protects cytokine-induced cytotoxicity and MHC class II antigen expression in vitro in the continuously growing and differentiated thyroid cell line, FRTL (Shon et al., 2004). GGT shows no evidence of general immunosuppression and the convenience of oral administration, has made it a drug of considerable interest in the treatment of autoimmune disorders.

Glycyrrhizae Radix has been popularly used as one of the oldest and most frequently employed botanicals in herbal medicine in Asian countries. Previously, G. radix extract (GRE) and its representative active components prevented both apoptotic and non-apoptotic cell injury (Kim et al., 2004a,b). In addition, GRE was reported to inhibit mitochondrial Bad translocation and caused restoration of mitochondrial Bcl(xL) and cytochrome *c* levels. Among the major components present in GRE, liquiritigenin, liquiritin, isoliquiritigenin or glycyrrhizin, exerted cytoprotective effect. The water extract of G. radix contained high levels of liquiritin, liquiritin apioside, isoliquiritin, isoliquiritin apioside and glycyrrhizin. Liquiritin and liquiritin apiosid and liquiritin apioside dose-dependently inhibited the number of coughs and methysergide, a serotonin receptor antagonist, antagonized the antitussive effect of liquiritin apioside (Kamei et al., 2003). Black soybean has also been used as a health food and herb in China and Korea for hundreds of years. Oral administration of polysaccharide isolated from Black bean in mice not only restored the leukocyte counts reduced but also enhanced granulocyte colony formation of bone marrow cells, confirming that polysaccharide promotes myelopoiesis activity in the bone marrow, stimulates production of various hematopoietic growth factors from spleen cells, and reconstitutes bone marrow (Liao et al., 2005). Also, the purified polysaccharide inhibited proliferation and induces differentiation of human leukemic U937 cells via activation of mononuclear cells. The molecular weight of polysaccharide component was about 480,000 by gel filtration. Structural analysis revealed that (1,6)-α-D-glucan was its major active component (Liao et al., 2001).

On the other hand, Angelicae Radix was known to have the protective effect against the toxicity among the ingredients. The water extract contained sodium L-malate, which was found to exhibit protective effects against both nephrotoxicity and bone marrow toxicity (Sugiyama et al., 1994). Ferulic acid, ligustilide and adenosine as the constituents were also reported (Guo et al., 2003; Song et al., 2004). For Cnidii Rhizoma, four alkylphthalides, cnidilide, (*Z*)-ligustilide, (3*S*)-butylphthalide, and neocnidilide as main constituents with acetylcholinesterase inhibitory activity were reported (Tsukamoto et al., 2005). Ethyl acetate soluble fraction of *Cnidium officinale* MAKINO and its constituents of *N*-methyl-β-carboline-3-carboxamide and amines inhibited neuronal cell death by reduction of excessive nitric oxide production in lipopolysaccharide-treated rat hippocampal slice cultures and microglia cells (Kim et al., 2003).

While the mechanism of action of GGT is basically unknown, it is well documented that it does not induce an overt inhibition of immunological functions. GGT treatment reduced the in vitro formation of adherent macrophage colonies and directly interfered with antigen presentation to T-cell lines, probably by inducing overactivation and apoptosis of macrophages. The importance of the balance between lymphocyte subpopulations, especially the Th1/Th2 ratio, in the development of autoimmunity (as in the case of multiple sclerosis) has been emphasized in several studies (Chen et al., 1994, 1996; Nicholson et al., 1995). In an effort to understand the mechanisms by which GGT downregulates autoreactivity, we investigated its effect on cytokine production, which reflects (corresponds to) the T-helper cell profile.

2. Materials and methods

2.1. Mice

Female SJL/J mice were purchased from Korea Research Institute of Bioscience and Biotechnology (Daejon, Korea) and housed in top-filtered cages in a standard animal facility. All mice were 6–8 weeks of age at the time of use. Animals were fed a regular diet and given water ad libitum without antibiotics.

2.2. Materials

Con-A Sepharose (particle size: 45–165 μm, 10–16 mg Con-A per gel sediment) was obtained from Pharmacia (Uppsala, Sweden). Calibration proteins for molecular weight determination were obtained from Pharmacia. pTg was from Sigma Chemical Co. (St. Louis, MO, USA) and other chemicals were of analytical reagent grade and obtained from Merck (Darmstadt, F.R.G).

2.3. Cytokine production in vitro and enzyme-linked immunosorbent assays

Single cell suspensions of lymphocytes were prepared from spleens obtained from naive SJL/J mice and from mice orally treated with GGT (80 mg/(kg day)) at 1, 2, and 5 days of treatment. The lymphocytes were incubated in vitro in 24-well plates in a standard proliferation medium (Ben-Nun et al., 1981), without, or with concanavalin A (Con A) (1 μg/ml). Aliquots of supernatants (0.02 ml) were collected at 12, 24, 48 and 72 h following incubation. The levels of the cytokines interleukin (IL)-4, IL-10 and interferon-γ (IFN-γ) were evaluated using commercial ELISA kits (Immunotech). Recombinant cytokines (Immunotech) were used for titration. Anti-thyroglobulin autoantibodies were assayed using standard ELISA methods (Engvall and Perlmann, 1971). The plates were coated overnight at 4 °C with mouse thyroglobulin at 10 μg/ml. A goat anti-mouse polyvalent immunoglobulin (IgM, IgG and IgA) conjugated to alkaline phosphatase (Sigma Immunochemicals) was used as developing antibody. The values shown are the mean optical density readings at 410 nm ± S.E. Isotypes were evaluated by using anti-mouse IgG1, -2a or -2b at 1/1500 (Sigma).

2.4. Detection of the mRNA of cytokines by polymerase chain reaction

Splenocytes from naive/untreated mice and mice treated for 1, 2 or 5 days with GGT, were obtained. RNA was extracted from the lymphocytes, before, or after 6 h of in vitro stimulation with ConA, and the mRNA of IL-4, IL-6, IL-10 and IFN-γ cytokines, as well as for transforming growth factor-β (TGF-β) and IL-12 was detected by a semiquantitative reverse transcriptase polymerase chain reaction (RT-PCR) method, as described previously (Cua et al., 1995).

2.5. Preparation of GGT

GGT was dissolved in tap water at a concentration of 200 mg/l to obtain an approximate dose of 100 mg/(kg day). The pH was then adjusted to 7.0. Controls were given normal tap water. GGT was added in vitro at 100 μg/ml (Cho et al., 1999).

2.6. Preparation of dgpTg

Porcine Tg (pTg) from Sigma (St. Loius, MO, USA) was diluted in distilled water and denatured by boiling in a microwave oven until a visible precipitation was observed. Deglycosylated porcine Tg (dgpTg) was characterized by chromatography and electrophoresis (Fig. 1), as described (Kim et al., 2004a,b). Native pTgor dgpTg (50 μg) was electrophoresed on a 5–25% (w/v) acrylamide gel and 0.1% (w/v) SDS with a Tris–glycine buffer system by the method of Laemmli (1970). Twenty milligrams of dgpTg or pTg in 500 μl distilled water were loaded onto Sepharose 4B gel; whereas pTg eluted as two peaks of 660 and 330 kDa, respectively, dgpTg eluted as two peaks of about 600 and 300 kDa. SDS-PAGE of pTg also showed a band at 330 kDa, whereas most of dgpTg was located in a band at 250 kDa (Fig. 1). These data suggested that deglycosylation induced aggregation of pTg.

After electrophoresis, the gel was stained with coommassie blue. After endo-β-*N*-acetylglucosaminidase F (Endo F) and trifluoromethane sulfonic acid (TFMS) experiments, SDS-PAGE were loaded with protein amounts allowing only the visualization, after Coomassie staining, of the major deglycosylated Tg. This avoided difficulties of interpretation due to the presence of minor Tg forms, which are not fully deglycosylated. Electrophoresis of dgpTg showed two bands at 330 and 660 kDa, whereas most of native pTg was located in a band at 660 kDa (Kim et al., 2004a,b). Digestion of the pTg by Endo F yielded one reaction product to the glycosidase action. The product

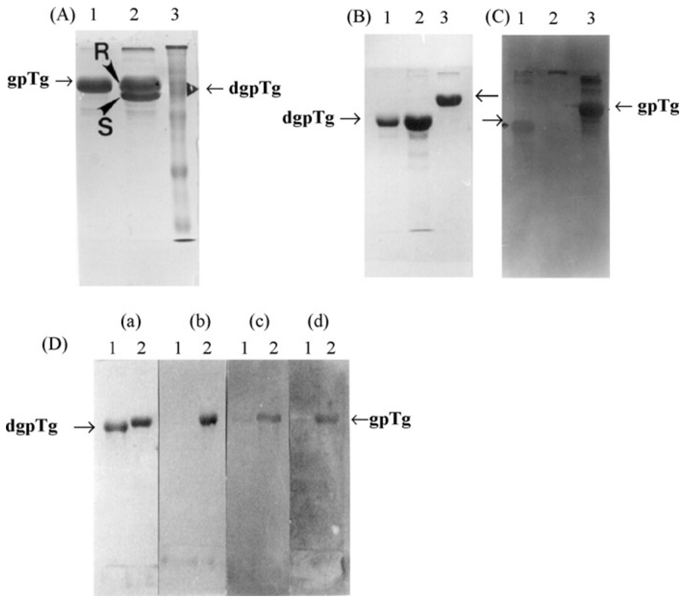

Fig. 1. (A) SDS-PAGE analysis of the purified pTg, after Endo F digestion (lane 2) in comparison with the untreated pTg (lane 1). Lane 3, standard molecular weight. R, resistant protein (about 330 kDa) against Endo F digestion; S, sensitive proteins (about 250 kDa) against Endo F digestion allowing effective deglycosylation. Glycan detection of the pTg. The pTg was separated on 6% SDS-PAGE, transferred to nitrocellulose membrane and stained for glycan by DIG Glycan Detection Kit (BM). (B) Coomassie staining for protein. (C) Glycan staining by lectin-labeled glycan detection kit. Lane 1, transferrin (25 μg) as a positive control; lane 2, bovine serum albumin as a negative control; lane 3, pTg. Arrowheads indicate the standard glycoprotein. (D) SDS-PAGE analysis of the pTg, after chemical deglycosylationwith trifluoromethanesulfonic acid (TFMS) and affinoblotting analysis using Con A and WGA-labeled digoxigenin. (a) Coomassie staining. (b) Glycan staining by digoxigenin detection kit (BM). (c) Con A-specific glycan staining using Con A-labeled digoxigenin kit (BM). (d) WGA-specific glycan staining using WGA-labeled digoxigenin kit. Lane 1, after deglycosylation; lane 2, before deglycosylation. Arrowhead indicates the position of glycosylated Tg.

differed from the original pTg in molecular weight by about 8000 Da. This, in conjunction with the experiment of Cona and WGA binding indicated the presence of high mannose N-linked oligosaccharides in the pTg.

Before Con A or WGA/alkaline phosphatase staining, proteins were transferred from SDS-gels to nitrocellulose paper as previously described (Kim et al., 1990). N-linked oligosaccharides were characterized by digoxigenin specific antibody conjugated with alkaline phosphatase (DIG Glycan detection Kit, Boehringer Mannheim, GmbH, Germany) and affinoblotting with Con A/Alkaline phosphatase (Lectin Con A-biotin-labeled, BM) or WGA/alkaline phosphatase (Lectin WGA-biotin-labeled, BM) as described by Hawkes (1982), but with skim with milk as a blocking agent. Enzymatic removal of the high mannose glycans with Endo-F was performed according to Trimble and Maley (1984). Tg (2 mg protein) was deglycosylated by 5.0 units Endo F (Boehringer Mannheim) in 400 μl of 25 mM acetate buffer (pH 5.0) at 37 °C for 12 h. Deglycosylation with TFMS was carried out according to Edge et al. (1981). The molecular weight of deglycosylated Tg was estimated by FPLC Sepharose-4B gel filtration and SDS-PAGE.

2.7. *Affinity chromatography on Con-A-sepharose 4B*

Affinity chromatography on native and deglycosylated Tg was performed with 0.5 ml (6.5 mm × 10 mm i.d.) column of Con-A-Sepharose 4B at 4 °C. The column was packed by sedimentation. The absorption capacity of the column for Tg was determined as follows: the affinity column was equilibrated in phosphate buffer (20 mM sodium phosphate, 150 mM NaCl, 0.02% NaN_3, pH 6.5) and the test solution consisting of 5 mg/ml of the Tg-containing solution in Tris–HCl buffer as above. Portions each of 0.5 ml were applied on the column in such a manner that as soon as the first aliquot had just entered into the column, the next portion was applied. Fractions (0.5 ml) were collected at a flow rate of 2 ml/min and 20 μl of every third fraction was subjected to SDS-PAGE. After having applied a total 50 mg (10 ml) Tg solution on the column, the remaining unbound and non-specifically bound material was eluted with 10 ml phosphate buffer (as above) and 10 ml phosphate-buffer–ethylene glycol (20% ethylene glycol in phosphate buffer) at a flow rate of 5 ml/h. After removal of residual ethylene glycol from the column by washing with 5 ml phosphate buffer, specifically bound material was eluted sequentially at a flow rate of approximately 5 ml/h with 5 ml 0.5 M α-methyl mannoside in phosphate buffer. These were subjected to affinoblotting using Con-A and WGA for glycosylated Tg.

2.8. *Immunization with Tg and dgpTg*

Injections were performed i.v. on days 0 and 14, with a dgpTg (100 μg) in saline without adjuvant. Simultaneously, pTg was emulsified in complete freund adjuvant (CFA) for immunization on day 0 and in incomplete freund adjuvant (IFA) for challenge on day 14. CFA suspension, which contained 1 mg/ml *Mycobacterium tuberculosis* (Difco Lab, Detroit, MI, USA), was injected s.c. with a pTg (100 μg). Control mice were injected with adjuvant suspensions alone (s.c) or saline (i.v.).

2.9. *Statistics*

Student's *t*-test was used where appropriate.

3. Results

3.1. *Characterization of dgpTg: glycan detection and affinity for Con A and WGA*

dgpTg was characterized by chromatography and electrophoresis. Twenty milligrams of dgpTg or pTg in 500 μl distilled water were loaded onto Sepharose 4B gel; whereas pTg eluted as two peaks of 660 and 330 kDa, respectively, dgpTg eluted as two peaks of about 600 and 300 kDa. SDS-PAGE of pTg also showed a band at 330 kDa, whereas most of dgpTg was located in a band at 250 kDa (Fig. 1A). These data suggested that deglycosylation induced aggregation of pTg. Endo F is known to specifically remove the high-mannose N-linked oligosaccharides. After enzymatic digestion of the pTg, products were analyzed by SDS-PAGE. The molecular weight was noted for the pTg (Fig. 1A). Digestion of the pTg by Endo-F yielded one reaction product to the glycosidase action. The product differed from the original pTg in molecular weight by about 8000 Da. This, in conjunction with the experiment of Con A and WGA binding indicated the presence of high mannose *N*-linked oligosaccharides in the pTg.

After separation of the pTg on 6.0% SDS-PAGE, glycan portion of glycoprotein was specifically detected using a digoxygenin specific antibody conjugated with alkaline phosphatase. As expected, pTg was positively stained with the glycan detection kit (Fig. 1B and C) when transferrin as a positive control was stained with positive signal (lane 1) and bovine albumin as a negative control was not stained. Not only high-mannose oligosaccharides sensitive to Endo-F, but also some complex side chains resistant to Endo-F were shown to interact with Con A. However, differential affinity of Con A for glycoproteins is well known, and the presence of high mannose oligosaccharides which poorly bind to Con A cannot be completely ruled out. Con-A/alkaline phosphatase or WGA/alkaline phosphatase affinoblotting after SDS-PAGE of the pTG and dgpTg was used to detect *N*-linked oligosaccharides which bind to Con A. Results showed that the pTg was positively stained with Con A and WGA while deglycosylated pTg was not stained with any reagents of glycan detection (Fig. 1D). The TFMS chemical was shown to efficiently remove glycans without peptide backbone cleavage. By SDS-PAGE analysis after TFMS treatment of the pTg, one reaction product was observed (Fig. 1D). The molecular weight differed from the native pTg by about 5000 Da. The molecular weight of the TFMS-treated polypeptide is similar to the molecular weight of the Endo F-digested pTg and it is possible that this form contained only high-mannose glycans.

3.2. Increased IL-4 and IL-10 production following treatment with GGT

As seen in Figs. 2 and 3, lymphocytes obtained from GGT-treated mice secreted increased amounts of IL-4 and IL-10 after in vitro stimulation with ConA. The increase in IL-10 was observed after 12 and 24 h of in vitro stimulation with ConA, and was most prominent following 5 days of treatment with GGT (2× increase compared to baseline – before treatment – values) (Fig. 2). Upregulation of IL-4 was more pronounced and peaked on day 10 of treatment with GGT (40× increase compared to baseline levels) (Fig. 3). On the other hand, secretion of IFN was not significantly affected following treatment with GGT (Fig. 4).

3.3. Cytokine mRNA detection by PCR

RT-PCR results showed an increased mRNA production of the cytokines IL-4 and IL-10 in the mice treated with GGT. The peak mRNA production for both these cytokines was observed following 5 days of treatment (Fig. 5). The earlier peak, as compared to that noted in the ELISA assay (peak on days 5–10), can be explained by the fact that probably mRNA production for these cytokines is upregulated earlier and the increased secretion follows later. This increased mRNA production was mainly detected when lymphocytes were examined immediately after removing them from the treated animals, without any stimulation in vitro; following in vitro stimulation of lymphocytes for 5 h with Con A the changes were not easily observed, since even naive lymphocytes (day 0 of treatment) had significant baseline mRNA levels for these cytokines and therefore the sensitivity of the test was too low to detect small changes. The finding of increased ex vivo production of mRNA without in vitro stimulation indicates that, already in vivo, GGT had induced an overactivation state of lymphocytes for the TH2 cytokines

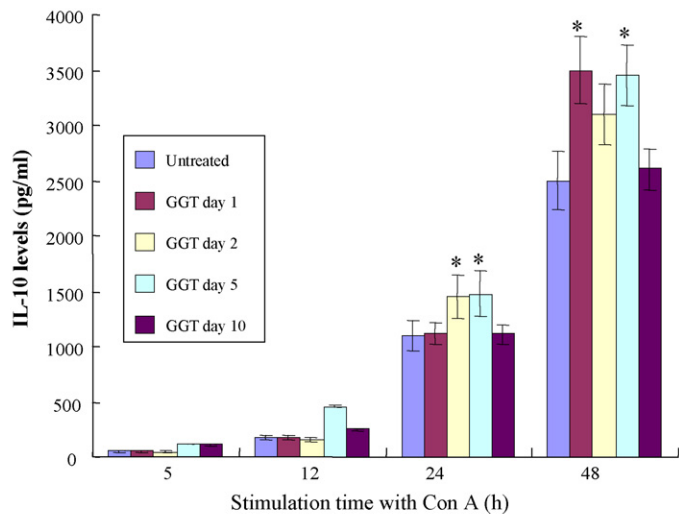

Fig. 2. GGT treatment upregulates IL-10 production by lymphocytes. Splenocytes were obtained from naive/untreated SJL/J mice and from mice treated with GGT for 1, 2, 5 or 10 days, and were stimulated in vitro with ConA for various periods (5, 12, 24 or 48 h). The levels (pg/ml) of IL-10 were examined by ELISA kits in the supernatants of the cultures. Asterisk (*) denotes statistically significant differences (at $P < 0.001$, two-tail *t*-test) between IL-10 levels produced by lymphocytes from untreated animals as compared to lymphocytes from GGT-treated mice.

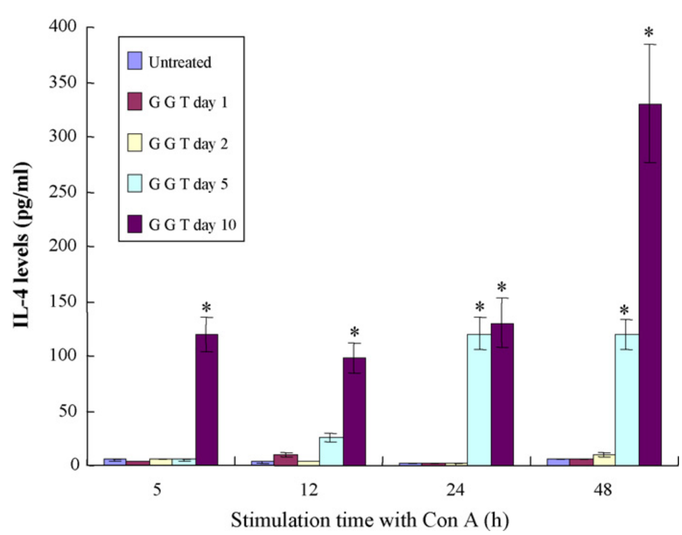

Fig. 3. GGT treatment upregulates IL-4 production by lymphocytes. Splenocytes were obtained from untreated SJL/J mice and from mice treated with GGT for 1, 2, 5 or 10 days, and were stimulated in vitro with ConA for various periods (5, 12, 24 or 48 h). The levels (pg/ml) of IL-4 were examined by ELISA kits in the supernatants of the cultures. Asterisk (*) denotes statistically significant differences (at $P < 0.001$, two-tail *t*-test) between IL-4 levels produced by lymphocytes from untreated animals as compared to lymphocytes from GGT-treated mice.

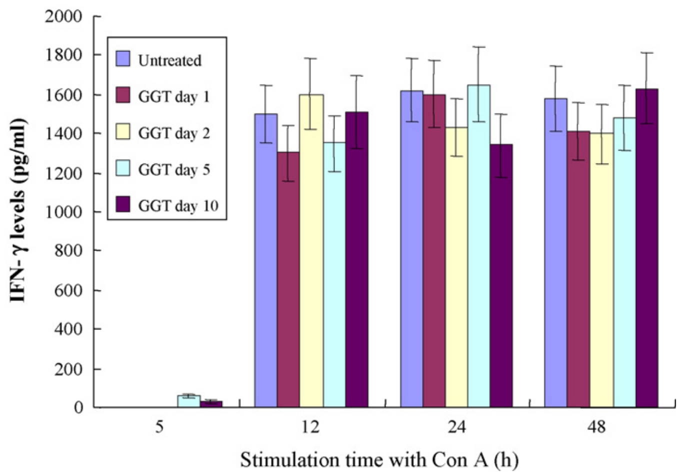

Fig. 4. GGT treatment does not significantly affect IFN production by lymphocytes. Splenocytes were obtained from untreated SJL/J mice and from mice treated with GGT for 1, 2, 5 or 10 days, and were stimulated in vitro with ConA for various periods (5, 12, 24 or 48 h). The levels (pg/ml) of IFN-γ were examined by ELISA kits in the supernatants of the cultures.

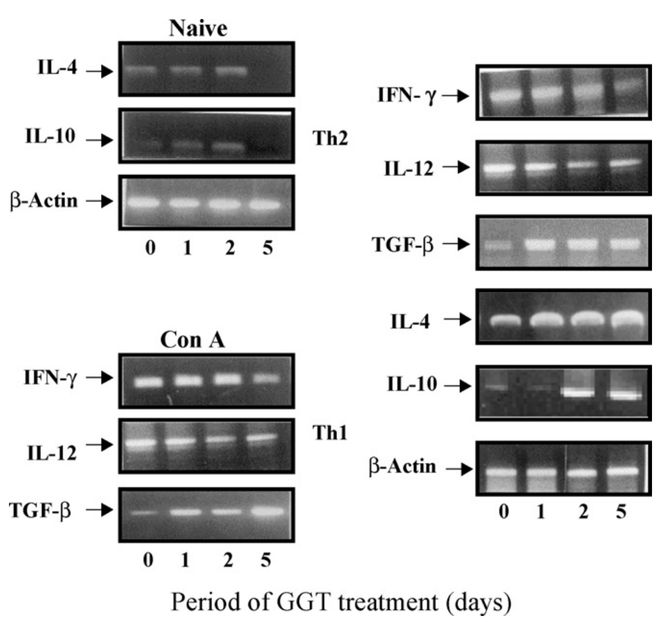

Fig. 5. GGT treatment upregulates TH2 and down-regulates TH1 cytokine mRNA production by lymphocytes. Splenocytes were obtained from naive/untreated SJL/J mice and from mice treated with GGT for 1, 2 or 5 days, and were stimulated in vitro with ConA for 5 h. The quantities of mRNA for IL-4, IL-10, IL-12, TGF-β and IFN-γ were evaluated by a semiquantitative RT-PCR method.

(in vivo TH2 shift). mRNA production of the proinflammatory cytokines, IL-12 (on days 2 and 5) and IFN-γ (on day 5), following 6 h of in vitro stimulation with ConA, was down-regulated by GGT treatment (Fig. 4), whereas the mRNA for TGF-another anti-inflammatory cytokine-was increased by GGT (on day 5).

4. Discussion and conclusion

Most existing treatments for autoimmunity are based on non-specific immunosuppression or anti-inflammatory agents and although often very effective at down-regulating the relevant autoimmune responses, may also lead to increased susceptibility to other complications associated with depressed immune function. EAT was induced by i.v. injection of deglycosylated senogeneic Tg without the aid of adjuvants. Denaturation of exogenous proteins prior to immunization was achieved by deglycosylation of pTg, which was demonstrated by electrophoresis, chromatography and glycan analysis using DIG-oxigenin labelling kit to induce its aggregation. The deglycosylated 660-kDa Tg was comparable to the 660-kDa globules of Tg isolated from bovine thyroid glands (Herzog et al., 1992), which represents a form of storage of highly concentrated Tg (100–400 mg/ml). Release of such Tg globules from normal thyroid glands of rats was observed in vivo, following specific signals for exocytosis (Ericson et al., 1983).

In the present study we have shown that treatment of SJL/J mice with GGT induces a shift of the lymphocytes towards the Th2 phenotype, as evidenced by their cytokine secretion profile. We have shown both by ELISA and by semiquantitative RT-PCR methods that, following treatment with GGT, spleen cells produce increased quantities of IL-4, IL-10 and TGF-β, whereas the secretion of Th1 cytokines, IFN-γ and IL-12, is either unchanged or down-regulated.

By antigen presentation of macrophages (APCs), lymphocytes (mainly of the helper CD4 subtype) expressing the specific T-cell receptor (TCR) for the antigen further differentiate, becoming either Th1 cells, that produce IL-2, IL-12, TNF-α and IFN-γ (pro-inflammatory cytokines), or Th2 cells, secreting IL-4, IL-6; TGF-β and IL-10 (Cua et al., 1995; Khoruts et al., 1995; Khoury et al., 1995). There is a cyclic relation between Th1 and Th2 lymphocytes. Under certain conditions Th1 cells acquire a Th2 phenotype and vice versa. In EAT and multiple sclerosis (MS), the lymphocytes involved in the inflammatory process are mainly of the Th1 phenotype (Issazadeh et al., 1996). Therefore, a ‘shift’ towards the Th2 subtype may lead to down-regulation of the Th1 lymphocytes, resulting in attenuation of inflammation. Such a shift can be achieved by oral tolerization techniques (Chen et al., 1994; Weiner et al., 1994) by cytokines such as IFN-γ, IL-4 and IL-10 (Porrini et al., 1995; Crisi et al., 1995) and, as shown in this study, by GGT. In addition, IL-4 and IL-10 have been shown to induce TH2 responses and to be effective in the treatment of EAT (Crisi et al., 1995).

Recently, some of the basic immunological concepts for self-/non-self discrimination have been revised (Forsthuber et al., 1996). It seems that inside the lymphocytic ‘arsenal’ even cells with an autoimmune potential exist, and under certain circumstances can expand and induce autoimmunity. However, several control mechanisms prevent the expansion of such autoimmune clones. All these immune populations are in a complex relation and interaction within the immune network. In addition, it seems that the acquisition of a specific phenotype is rather transient and the status of lymphocytes can be changed according to the immunological milieu. So, there are no ‘good’ and ‘bad’ cells but the same cell can behave as a Th1 or a Th2 lymphocyte, as a ‘memory’ helper cell or as a suppressor-inducer, under certain circumstances. The shift from Th1 to Th2 phenotype may be a transient event and is not universally helpful in autoimmunity (Mosmann and Sad, 1996). Our results show that such a shift towards Th2 (which was induced by GGT) may be helpful in T-cell-mediated autoimmune diseases like EAT and MS, but such a shift should not be permanently long-lasting since it may be associated with increased danger for induction of antibody-mediated autoimmunity.

GGT was shown, in our previous work, to be one of the most potent agents ever used to treat EAT (Han et al., 1999). According to the data presented here, it seems that one of the main mechanisms involved in down-regulation of EAT, by GGT, is the Th1 to Th2 shift. This is in accordance with a recent study, which showed that addition of GGT in vitro, in human monocytes, down-regulates Th1 cytokine production (Engvall et al., 1971). Such a therapeutic approach with immunomodulation and stimulation of down-regulatory naturally existing mechanisms, rather than generalized immunosuppression, seems to offer great advantages for the management of autoreactivity, and may represent the mainstay of future treatments for MS and other T-cell-mediated autoimmune diseases. GGT consisted of four medicinal herbs of *Glyrrhiza uralensis* Fischer, *Glycine max* L. Merr, *Angelica gigas* Nakai and *Cnidium officinale* Makino, and it has long been used for the prevention of cancer metastases (Cho et al., 1999) and for the autoimmune diseases, including EAT and collagen-induced arthritis (CIA) (Han et al., 1999). GGT has also potent inhibitory effects on human A549 lung carcinoma cells, cervix epithelioid carcinoma Hela cells and Hepa1c1c7 hepatoma cells (Cho et al., 2000a,b). In addition, GGT increased the immune-stimulating activities on T and B cells with cytolytic activity against K562 cells, indicating its immunomodulating effect (Cho et al., 2000a,b). Therefore, GGT has shown exceptional promise as an immunomodulator having no generally immunosuppressive effects but inducing marked amelioration of several autoimmune diseases. Furthermore, it protects cytokine-induced cytotoxicity and MHC class II antigen expression in vitro in the thyroid cells (Shon et al., 2004). GGT shows no evidence of general immunosuppression and the convenience of oral administration has made it a drug of considerable interest in the treatment of autoimmune disorders. Furthermore, GGT prevented the experimental thyroiditis induced by the immunization of CBA/J mice (H-2^k) with deglycosylated pig thyroglobulin using two separate adjuvants (CFA and LPS) (Han et al., 1999).

In conclusion, our data indicate that intervention therapy although promising, should be approached cautiously since a knowledge of the underlying mechanism may be crucial for predicting the success or otherwise of GGT treatment under different conditions.

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